

An explicit 5×4 square-sum specialization and a genus-13 reduction

Yicheng Pan
18-year-old independent researcher
a0917212213@gmail.com

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Abstract

We give an explicit pair of integer sets of cardinalities five and four whose twenty pairwise sums are squares. The example arises from a one-parameter subfamily of Choudhry’s rational 5×3 construction. An elliptic-curve addition produces a candidate fourth shift as a rational function of the parameter; the two remaining conditions reduce to the simultaneous rational-square values of two explicit degree-eight polynomials. Their fiber product has genus 13, and their difference admits a useful factorization. The specialization $m = -8/19$ satisfies both conditions. We also give an exact finite certificate that the resulting five-column set has no further integer shift. For rational extensions of the fixed grid, a complete two-square parametrization gives natural genus-three necessary covers. Bounded rational-point searches find only the displayed rows and columns, while all thirty-six natural covers have Jacobian rank above their genus. This is an auxiliary square-sum result and does not solve Erdős Problem 885 for $k = 5$.

1 Introduction

For finite sets $A, B \subset \mathbb{Z}$, call (A, B) a square-sum pair if $a + b$ is a square for every $a \in A$ and $b \in B$. Choudhry [1] constructed multiparameter examples of sizes 3×3 , 5×3 , and 4×4 , and asked for examples of sizes 5×4 and 5×5 . Mausberg subsequently posted a 5×4 square packet in connection with Erdős Problem 885 [5]. The present note records a distinct 5×4 specialization and, more importantly, an explicit algebraic reduction that explains how it arises inside Choudhry’s family.

The relation with the factor-difference problem is

$$d \in D(N) \iff d^2 + 4N \text{ is a square.}$$

Thus five positive integers with five common factor differences would give a 5×5 square-sum grid; after adjoining the zero row, this is an anchored 6×5 grid. The construction below has only four anchored rows, so it does not settle the $k = 5$ case.

2 A one-parameter subfamily

In Choudhry’s 5×3 family, set

$$(f, g, u_1, u_2, v_1, v_2) = (4, 1, 2, 1, 3, 1)$$

and retain $m \in \mathbb{Q}$ as a parameter. The five columns are

$$\begin{aligned} a_1 &= 4(24m^2 + 145m - 4)^2, \\ a_2 &= 100(24m^2 + 49m + 4)^2, \\ a_3 &= 25(97m^2 + 8)^2, \\ a_4 &= 4(24m^2 - 145m - 4)^2, \\ a_5 &= 100(24m^2 - 49m + 4)^2, \end{aligned}$$

and three shifts are

$$\begin{aligned} b_1 &= 0, \\ b_2 &= 96(3m - 4)(3m + 4)(8m - 1)(8m + 1), \\ b_3 &= 3300(m - 1)(m + 1)(6m - 1)(6m + 1). \end{aligned}$$

All fifteen sums $a_i + b_j$ are rational squares.

Put

$$D = 97m^2 - 8m - 8$$

and, for $D \neq 0$, define

$$b_4 = \frac{12(2m - 1)(2m + 1)(3m - 1)(3m + 1)}{D^2} \times (35m + 4)(85m - 28)(89m - 20)(109m + 20). \quad (1)$$

The expression (1) was found by elliptic-curve addition on the curve determined by the columns (a_1, a_3, a_5) . No sign convention for that group law is needed below: the following proposition is proved by direct identities.

Proposition 1. *For b_4 as in (1), the sums $a_1 + b_4$, $a_3 + b_4$, and $a_5 + b_4$ are rational squares. The remaining two sums are rational squares if and only if $F_2(m)$ and $F_4(m)$ below are rational squares:*

$$\begin{aligned} F_2(m) &= 3252420900m^8 - 268152720m^7 - 846520196m^6 \\ &\quad + 226981960m^5 + 70762193m^4 - 37069520m^3 \\ &\quad - 3000736m^2 + 1580800m + 160000, \\ F_4(m) &= 3122350884m^8 - 865433712m^7 - 1137811292m^6 \\ &\quad + 310045192m^5 + 136716497m^4 - 33456848m^3 \\ &\quad - 6350752m^2 + 978688m + 135424. \end{aligned}$$

Proof. Direct expansion gives

$$\begin{aligned} a_1 + b_4 &= \left(\frac{2(55878m^4 - 6572m^3 - 10761m^2 + 1128m + 368)}{D} \right)^2, \\ a_3 + b_4 &= \left(\frac{121165m^4 - 14696m^3 - 23010m^2 + 2384m + 800}{D} \right)^2, \\ a_5 + b_4 &= \left(\frac{2(57030m^4 - 12052m^3 - 7095m^2 + 408m + 400)}{D} \right)^2, \end{aligned}$$

and

$$a_2 + b_4 = \frac{4F_2(m)}{D^2}, \quad a_4 + b_4 = \frac{4F_4(m)}{D^2}.$$

The assertion follows. $\square$

Proposition 1 replaces a five-column search by the simultaneous rational points on the two hyperelliptic curves $y^2 = F_2(m)$ and $z^2 = F_4(m)$.

Proposition 2. *The polynomials F_2 and F_4 are squarefree and coprime, and*

$$F_2 - F_4 = 24(m+4)(4m+1)(9m+4)(16m+1)D^2. \quad (2)$$

The smooth projective normalization of the fiber product

$$y^2 = F_2(m), \quad z^2 = F_4(m)$$

has genus 13.

Proof. Polynomial gcd calculations give $\gcd(F_i, F'_i) = 1$ and $\gcd(F_2, F_4) = 1$; direct expansion gives (2). Each double cover of the m -line is branched at the eight distinct finite roots of its octic, and the two branch loci are disjoint. For the resulting degree-four cover, each of the sixteen branch values has two points of ramification index two. Riemann–Hurwitz therefore gives

$$2g - 2 = 4(-2) + 16 \cdot 2 = 24,$$

so $g = 13$. □

3 The specialization $m = -8/19$

At $m = -8/19$ one has

$$F_2(m) = \left(\frac{35668080}{130321} \right)^2, \quad F_4(m) = \left(\frac{32892048}{130321} \right)^2,$$

and

$$b_4 = -\frac{15740769200}{1172889}.$$

Clearing denominators and translating the least shift to zero produces

$$\begin{aligned} A &= \{202720644, 827712900, 1476864900, 494706564, 80729856900\}, \\ B &= \{0, 358400700, 1398239500, 17139008700\}. \end{aligned}$$

Theorem 1. *Every element of $A + B$ is a square.*

Proof. The square roots, with rows indexed by the displayed order of A and columns by the displayed order of B , are

14238	23688	40012	131688
28770	34440	47180	134040
38430	42840	53620	136440
22242	29208	43508	132792
284130	284760	286580	312840.

Squaring the entries gives the twenty sums exactly. □

4 An exhaustive integer-shift certificate

The displayed grid cannot be extended by a fifth *integer* shift while keeping the same five columns.

Lemma 1. *Let $a < c$ be integers. Every integer b for which both $a + b$ and $c + b$ are squares arises from a factorization*

$$(x - y)(x + y) = c - a,$$

with $x, y \geq 0$, by $b = y^2 - a$.

Proof. Write $a + b = y^2$ and $c + b = x^2$ and subtract. □

For $a = 202720644$ and $c = 494706564$ the difference is

$$291985920 = 2^9 3^2 5 19 23 29.$$

There are 192 parity-compatible factor pairs with $x - y \leq x + y$. Testing the remaining three columns leaves exactly

$$0, \quad 358400700, \quad 1398239500, \quad 17139008700.$$

This is a finite proof, implemented independently of the parameter search. It does not exclude rational shifts with new denominators or different five-column sets.

The example is not affinely equivalent to the Mautsberg packet. Indeed, for ordered shifts $b_0 < b_1 < b_2 < b_3$ put

$$\lambda(B) = \frac{(b_0 - b_2)(b_1 - b_3)}{(b_0 - b_3)(b_1 - b_2)}.$$

Positive scaling and translation preserve this invariant. For the present shift set and the Mautsberg shift set, respectively, its values are

$$\frac{541359200}{411195147} \quad \text{and} \quad \frac{951142}{921557},$$

which are unequal.

5 Rational extensions of the fixed grid

There is a direct way to search for rational extensions without returning to the parameter m . Let $C_0, C_1 \in \mathbb{Q}$ be distinct, put $d = C_1 - C_0$, and suppose

$$C_0 + t = X^2, \quad C_1 + t = Y^2.$$

Writing $s = X + Y$ gives the complete parametrization

$$X = \frac{s - d/s}{2}, \quad Y = \frac{s + d/s}{2}, \quad t = \left(\frac{s - d/s}{2} \right)^2 - C_0. \quad (3)$$

Choose a known solution $s_0 \neq 0$, write $s = s_0 r$, and put

$$\rho = \frac{d}{s_0^2}, \quad Q_j(r) = r^4 + \left(\frac{4(C_j - C_0)}{s_0^2} - 2\rho \right) r^2 + \rho^2.$$

Then direct substitution in (3) gives

$$C_j + t = \frac{s_0^2 Q_j(r)}{4r^2}. \quad (4)$$

Consequently, if two further entries $C_a + t$ and $C_b + t$ are squares, then

$$w^2 = Q_a(r)Q_b(r). \quad (5)$$

Because the C_j are distinct, each quartic is squarefree and two different quartics are coprime. Thus (5) is a genus-three hyperelliptic curve. Searching the product may introduce points at which the two factors have the same nonsquare class; testing the individual factors in (4) removes these false positives and cannot remove a genuine extension.

For row extensions take $(C_j) = B$, the known value $t = 202720644$, and $s_0 = 14238 + 23688 = 37926$. For column extensions take $(C_j) = A$, the known value $t = 0$, and $s_0 = 14238 + 28770 = 43008$. Exact rational-point enumeration on the resulting degree-eight models through naive parameter height 10^6 returns respectively 42 and 34 affine points. Exact tests of every individual Q_j leave precisely the five displayed elements of A on the row side and the four displayed elements of B on the column side. These are bounded statements, not complete determinations of the rational points.

The polynomials in (5) are even and reciprocal under $r \mapsto \rho/r$. The two involutions split their Jacobians, up to isogeny, into three elliptic factors by the standard Kani–Rosen mechanism [6]. We exhausted all choices of the two entries used in (3) and the two remaining quartics used in (5): six models for row extension and thirty for column extension. Exact two-descent rank intervals give minimum total Jacobian ranks 13 and 11, respectively. Both exceed the genus, so none of these thirty-six natural models satisfies the classical Chabauty inequality. This explains why the most immediate completeness argument fails; it does not prove that the fixed grid has no rational extension. Independently of the rank program, a second implementation verifies all thirty-six squarefree models, the quotient maps and differential determinant, and the corresponding trace decomposition at three good primes per model.

As an independent bounded check, rank-three elliptic projections were searched in the coefficient cube $[-200, 200]^3$ for row extension and $[-250, 250]^3$ for column extension. The chosen bases were unchanged by saturation at primes below 1000. The cubes contain 64,481,201 and 125,751,501 representations, respectively; a twenty-prime sieve followed by exact rational-square tests found no new compatible value. These coefficient boxes are unrelated to the parameter-height bound above, and none is a global height bound.

6 Computational checks and bounded search

All symbolic identities in Proposition 1 were checked by exact polynomial cancellation. The twenty identities in Theorem 1 were checked by two independent integer programs. Reduced fractions $m = p/q$ with $0 < |p| \leq 20000$ and $1 \leq q \leq 20000$ were enumerated exactly. Homogeneous evaluations of F_2 and F_4 were first sieved modulo eight primes; every survivor was then tested by arbitrary-precision integer square root. Among 486,361,582 values there were nine simultaneous-square hits. Eight of these,

$$\pm \frac{1}{2}, \quad \pm \frac{1}{3}, \quad -\frac{4}{35}, \quad \frac{28}{85}, \quad \frac{20}{89}, \quad -\frac{20}{109},$$

make $b_4 = 0$. The only hit giving a new shift was $m = -8/19$. An independent Python enumeration agrees through height 2000. These are bounded computations, not a global uniqueness theorem.

Search code, identity certificates, and raw parameters are supplied with the companion verification archive. Computer algebra and AI-assisted exploration were used to locate and simplify the identities; every stated theorem reduces to the exact displayed formulas and integer checks.

7 Concluding remarks

The example answers the existence question for another 5×4 square-sum pair but does not address the more difficult 5×5 question, nor the anchored 6×5 configuration needed for Erdős Problem 885 at $k = 5$. The genus-13 curve in Proposition 2, together with the factorization (2), provides a compact starting point for determining whether this subfamily contains further 5×4 examples. For the fixed specialization, Section 5 shows that every natural genus-three extension projection has rank well above its genus. A global determination would therefore require a covering or combined Mordell–Weil sieve, quadratic-Chabauty input, or a different specialization with lower-rank quotients.

References

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